RAJALAKSHMI ENGINEERING COLLEGE

Fenulin E 251701012-2025-CSD F2

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|---|---|
| **Started on** | Monday, 28 September 2026, 9:36 PM |
| **State** | Finished |
| **Completed on** | Monday, 28 September 2026, 10:20 PM |
| **Time taken** | 44 mins 4 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
int count(int a[],int l,int r)
{
    if(l==r)
    {
        return a[l]==0;
    }
    int mid=(l+r)/2;
    return count(a,l,mid)+count(a,mid+1,r);
}
int main()
{
    int n,arr[100];
    scanf("%d",&n);
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    printf("%d",count(arr,0,n-1));
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |
| ✔ | 10<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1 | 0 | 0 | ✔ |

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 8<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0 | 8 | 8 | ✔ |
| ✔ | 17<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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